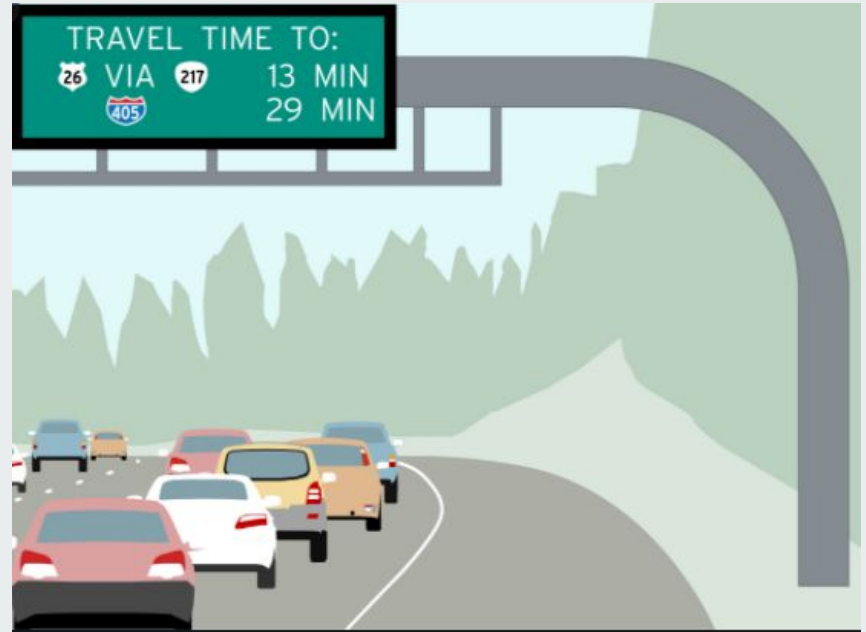
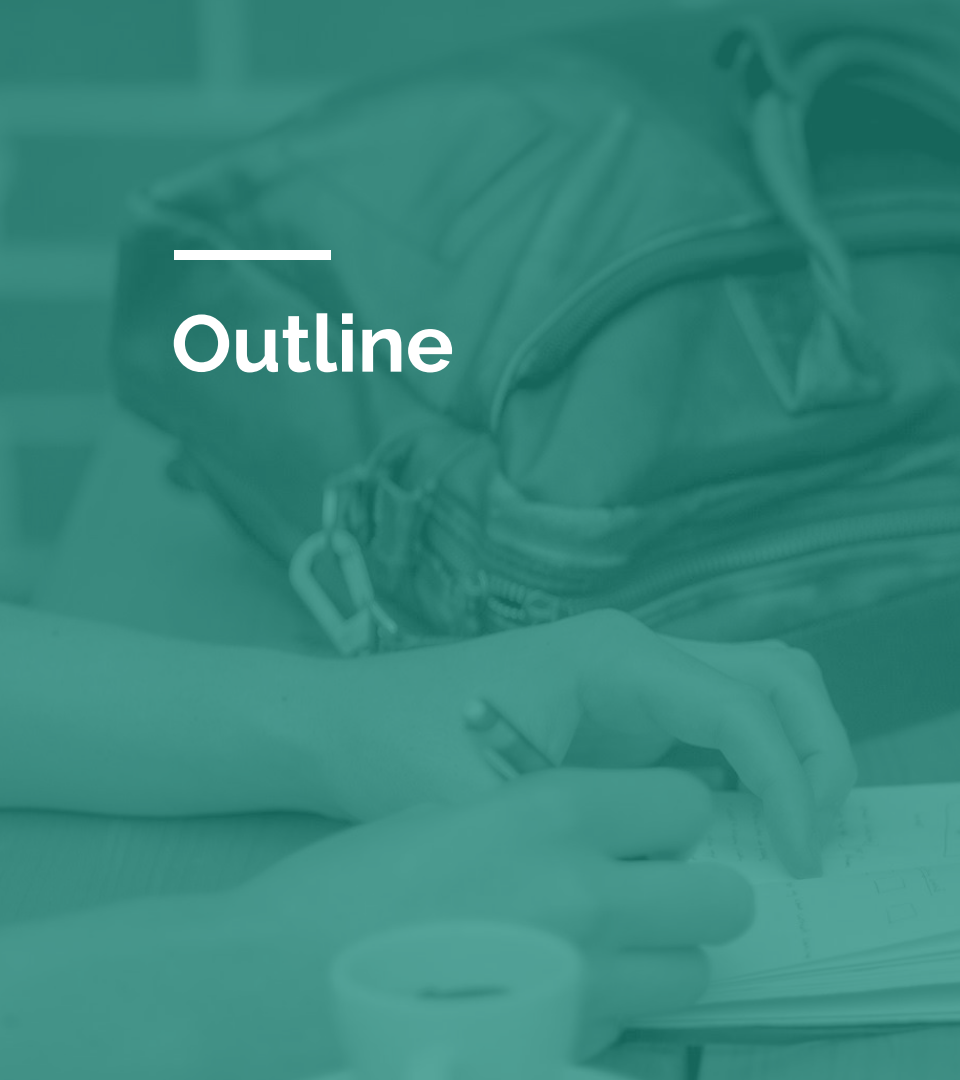


ANALYSIS OF TRAVEL TIMES

Making it there on time , all the time !





Outline

- INTRODUCTION
- TOOLS
- DATA COLLECTION
- CREATION OF CALCULATED FIELDS
- JOINING MULTIPLE TABLES
- FILTERS & MARKS CARD
- VISUALIZING DATA
- CONCLUSIONS



INTRODUCTION

- This study aims to analyse the impacts of crashes & severe weather on travel times.
- Develop travel time reliability measures such as Buffer Time that the travelers should add to their average travel time when planning trips to ensure on-time arrival.



TOOLS

- Tableau helps to transform data into actionable insights, thereby helps inform analysis & interpretation . It helps us to explore with limitless visual analytics, build dashboards and perform ad hoc analyses in just a few clicks. Share your work with anyone and make an impact .
- Tableau supports many functions for use in Tableau calculations.



Data Collection

- I collected data from the portal <http://portal.its.pdx.edu>. This is the official transportation data archive for the Portland-Vancouver metropolitan region. The data sets include Incident Data, Weather Data & Travel Time Data for the month of January, 2020. All the data is downloaded in CSV format.
- The important columns of Incident data are City, Create Time, Clear Time, County, Incident Type, Route Direction, Route Name, Max Lanes Affected.
- The important columns of Weather data are Report time, Weather text, station id (i.e., KPDX). Weather text is reported every hour throughout the day.
- The important columns of Travel Time Data are Start Date & Time, Average Travel Time. The average travel time is recorded with the resolution of 1 hr.



Creation Of Calculated Fields

- Rounding off create time of crash to nearest hour.

Hour Of Create Time Of crash

```
DATETIME(INT([Create Date & Time]) + ROUND(FLOAT([Create Date & Time])%1*24)/24)
```

- Rounding off Report time of weather to nearest hour. For example : 9:53 PM to 10:00 PM so that it can be compared with average travel time of 10:00 PM (which is average travel time of 9:00 PM to 10:00 PM)

Hour Of Report Time

```
DATETIME(INT([Report Date & Time]) + ROUND(FLOAT([Report Date & Time])%1*24)/24)
```

Creation Of Calculated Fields[Contd.]

- Duration of the incident

```
DATEDIFF('minute',[Create Date & Time],[Clear Time])
```

- Calculated field to highlight the hour of crash in visualizations.

Hour Of Crash

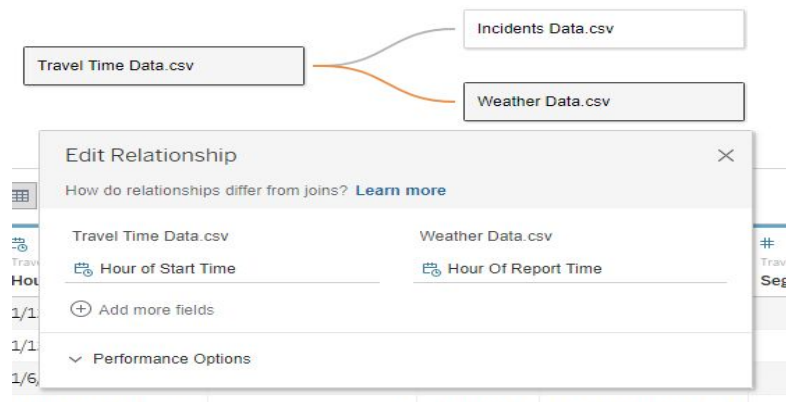
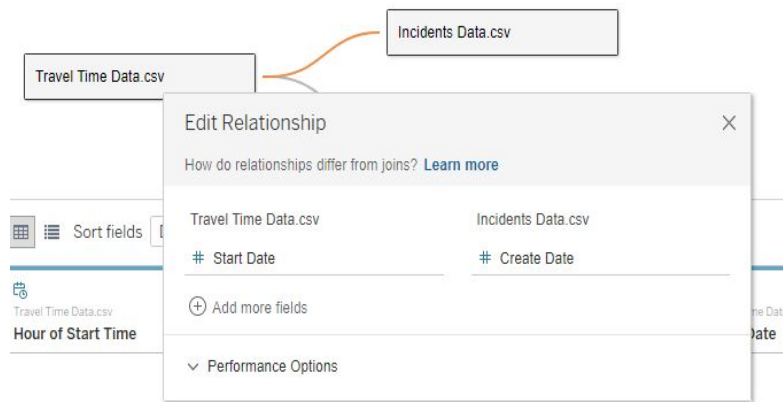
```
IF [Hour of Start Time]==[Hour Of Create Time Of Crash] THEN "Hour of crash"  
ELSE "NO crash"  
END
```

- Buffer Time

```
PERCENTILE([Average Travel Time],0.95)-AVG([Average Travel Time])
```

Joining Multiple Tables

- Split time & date field of each table into time field , date field for ease of analysis.
- Relationships are a flexible way to combine data for multi-table analysis in Tableau.
- Travel Data & Incident Data are related by Date fields.Travel Data & Weather Data are related by Time fields.



FILTERS & MARKS CARD

- You can use filters to clean up things that we don't need in our analysis. For example, for all the analysis related to crashes, I considered only Interstate-5 NB, Multnomah county.
- the **Marks card** provides you with control over how the data is displayed in the view.

Filters

Create Date

Incident Type: VH

Route Name: I-5

Route Direction: NB

County: Multnomah

City: Portland

Marks

▼ All

▼ AVG(Average Travel Time) ~

^ **CNT(Incidents Data.csv)** |||

||| Bar ▼

Color

Size

Label

Detail

Tooltip

MAX(Max Lanes Affected)



Visualizing Data : Welcome To My Dashboard !

- The drag and drop interface of tableau helps to perform tasks like sorting, comparing and analyzing, very easily and fast.
- Building dashboards with Tableau allows even non-technical users to create interactive visualizations. They can combine data sources, add filters, and drill down into specific information.
- We can publish our interactive data to web using tableau public . Anyone can download a copy of workbook with data & view how the visualizations are created. Here are my links to dashboards.
- Statistics Of Incidents :
<https://public.tableau.com/profile/sai.priyanka5815#!/vizhome/Analysisoftraveltime/StatisticsOfIncidents>
- Travel Time Analysis :
<https://public.tableau.com/profile/sai.priyanka5815#!/vizhome/Analysisoftraveltime/TravelTimeAnalysis>
- Weather Statistics :
<https://public.tableau.com/profile/sai.priyanka5815#!/vizhome/Analysisoftraveltime/WeatherStatistics>



CONCLUSIONS

- Multnomah county experience more number of incidents. Of all the routes, Interstate-5 has the highest number of incidents. Most number of incidents are vehicular incidents.
- Count of crashes in a day impact average travel time but this isn't the only factor that defines travel time. There are other maximum no. of lanes affected, duration of the incident, weather etc.
- Severe weather along with crashes worsens the travel time.
- Calculation of reliability metrics such as Buffer time helps us to budget extra time to arrive "On Time" for 95% percent of the trips.